ChemPubSoc Europe
DOI: 10.1002/ejic.201801512
EurJIC European Journal of Inorganic Chemistry
Minireview

Carbon Aerogels

Carbon Aerogels for Environmental Clean-Up

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Abstract: Carbon aerogels are a fascinating three-dimensional (3D) monolithic porous material with remarkable physicochemical properties, including low density, large surface area, abundant pore structure, high electrical conductivity, chemical stability, environmental compatibility, adjustable surface chemistry, as well as controllable structural features. These properties endow carbon aerogels with excellent adsorption and catalytic performance. Therefore, they are widely applied in environmental chemistry for removing pollutants like oils, toxic organic solvents, dyes, heavy metal ions in aquatic environments, and volatile organic compounds (VOCs), carbon oxides (CO_2, CO), nitrogen oxide (NO_x), and hydrogen sulfide (H_2S) in the atmosphere. In this paper, the preparation processes of carbon aerogels are fully reviewed, and their applications for environmental clean-up based on adsorption and catalytic action are thoroughly addressed. In each section, the most recent representative studies will be highlighted. Finally, the challenges, outlooks, and prospects in this emerging and promising research field will be briefly discussed.

1. Introduction

Carbon aerogels, the 3D porous network structure formed by carbon nanomaterials, have received widespread attention since they were firstly obtained[1,2] due to their fascinating physical properties, including low apparent density, large specific surface area, abundant pore structure, high electrical conductivity, good chemical stability, and environmental compatibility. Based on the above characteristics, a kaleidoscope of carbon aerogels has been fabricated and used widely in energy storage,[3,4] thermal insulation,[5] environmental clean-up,[6,7] chemical sensors,[8,9] and biomedical and pharmaceutical applications.[10,11] As shown in Figure 1, the number of publications concerning carbon aerogel have been increasing in the past decade. Following the breakthrough advance in the synthesis process since 2000, carbon aerogels have become a favored option which sufficiently meet most of the high-performance applications requirements. These advances even became more pronounced by coming out the new types of carbon aerogels during the last years.[12]

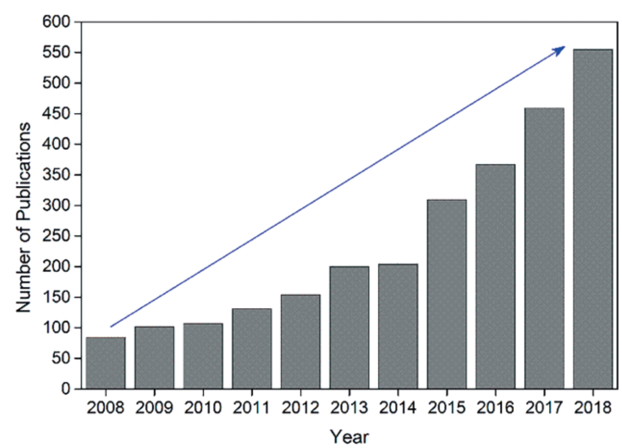

Figure 1. The growing number of publications containing “carbon aerogel”.

The preparation of carbon aerogels involves the following stages:[13] (I) gelation stage, the formation and reinforcement of the gel by sol-gel and aging processes; it is noteworthy that the aging process was regarded as one step of the gelation stage; (II) drying, the acquisition of aerogel by drying the gel under ambient, freeze or supercritical conditions; (III) carbonization, the formation of carbon aerogel by carbonizing aerogel at a high temperature under a flowing nitrogen atmosphere. At present, lots of carbon aerogels are commercially available which have been applied in real scenarios. According to the various sources of precursor, they are commonly divided into the following types: polymer carbon aerogels (PCAs),[14,15] graphene carbon aerogels (GCAs),[16,17] carbon nanotube carbon aerogels (CNCAs),[18,19] biomass carbon aerogels (BCAs),[20–22] and composite carbon aerogels (CCAs).[23,24] Different types of carbon aerogels have their own unique characteristics and are suitable for different applications.

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ORCID(s) from the author(s) for this article is/are available on the WWW under https://doi.org/10.1002/ejic.201801512.

Carbon aerogels show wonderful adsorption and catalytic performance because of the microscopic properties of carbon nanomaterials and the macroscopic structure of the gel. In addition, they are easy to recycle due to their macrostructure, super elasticity, and high-temperature resistance.[25] Therefore, their application for environmental clean-up has grown rapidly and

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drawn significant scientific interest.[26] Up to now, carbon aerogels have been used in lots of the remediation processes based on their adsorption and catalysis properties, like oil/water separation,[7,27,28] removal of heavy metal ions,[29,30] clean-up of volatile organic compounds (VOCs),[31] CO_2 capture,[32,33] removal of nitrogen oxide (NO_x)[34] and so on (Figure 2).

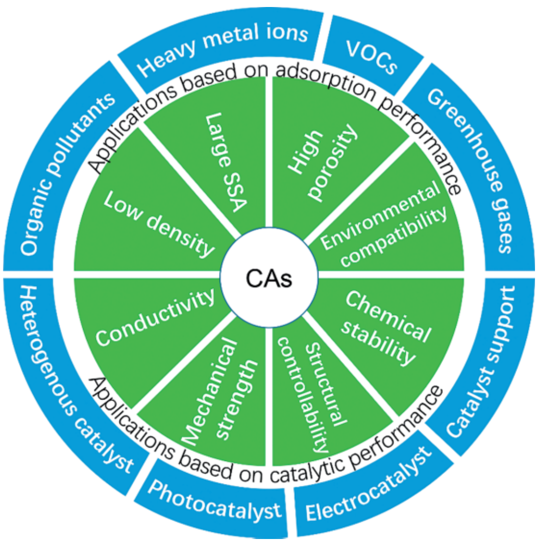

Figure 2. Schematic illustration of characteristics and applications of carbon aerogels.

To the best of our knowledge, there are only a few review articles related to carbon aerogels for the prospect of environmental clean-up.[2, 13, 26] This review aims at comprehensive and timely overview of the preparation of carbon aerogels as well as their applications in environmental protection field. Firstly, a brief description of the main stages of carbon aerogel preparation is presented. Subsequently, an overview of the recent development of carbon aerogels for environmental remediation based on adsorption and catalytic performance is presented. Finally, the challenges, outlooks, and prospects are provided.

2. Synthesis of Carbon Aerogels

Carbon aerogels were firstly prepared by polymerizing resorcinol with formaldehyde (RF) under alkaline conditions, followed with supercritical drying and carbonization.[1] Since then, more and more monomers were used, such as melamine-formaldehyde (MF),[35,36] phenol-formaldehyde (PF),[37] cresol-formaldehyde (CF),[38] phenol-furfural,[39] and some polymers, like polyacrylonitriles,[40,41] polystyrenes [42] and polyurethanes.[43] In the last few years, an increasing variety of carbon aerogels have emerged, like graphene aerogels,[14,40,44] carbon nanotube aerogels,[45,46] and biomass aerogels .[47–49] Generally, the preparation process for most of the carbon aerogels includes three stages: gelation, drying, and carbonization **(**Figure 3**)**, and each stage has a certain effect on the final properties.

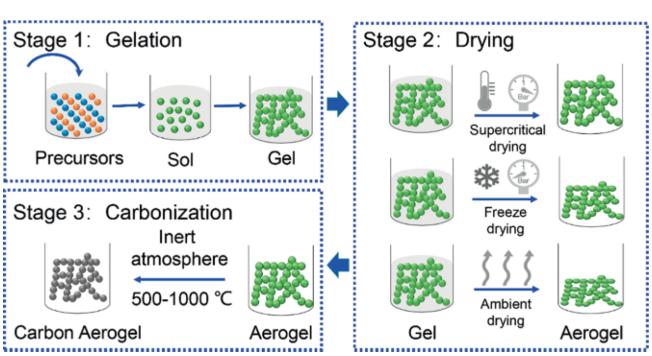

Figure 3. General preparation processes for carbon aerogels.

2.1 Gelation Stage

At the gelation stage, the hydrogel is formed by polymerizing and/or crosslinking of single precursor molecules. But the reactions are different for different precursors. For polymer aerogels, like resorcinol-formaldehyde aerogel, this stage includes four steps: (I) the formation of hydroxymethyl groups ($-CH_2OH$) by additional reaction derived from formaldehyde on resorcinol; (II) the polymerization of hydroxymethyl resorcinol through methylene ($-CH_2$) and ether–methylene ($-CH_2OCH_2$) bridge to form the clusters; (III) the formation of 3D hydrogel through the aggregation and crosslinking of the clusters; and (IV) aging, the further growth and strengthening of the as-prepared hydrogel with the unfinished chemical reactions.[1, 50, 51] At the beginning, alkali catalysts, including Na_2CO_3, K_2CO_3, $Ca(OH)_2$, etc., were used. Under alkaline conditions, resorcinol lost protons, forming the resorcinol anion, which had a strong nucleophilic addition capability influenced by the negative charge, and could easily react with formaldehyde to form hydroxymethyl resorcinol. And in recent years, acid catalysts were gradually applied. Unlike the situation under alkaline conditions, the acid catalyst system mainly increased the electrophilicity of formaldehyde, and the electrophilic addition of formaldehyde to resorcinol formed the hydroxymethyl resorcinol.

For other precursors, like graphene, CNTs, and biomass, the reaction at the gelation stage is simpler than that with polymers. Graphene sheets are reassembled by the interaction force, resulting in the 3D porous mesh structure. Up to now, there are a lot of methods devoted to the synthesis of graphene gels, such as the sol-gel method, template method, spacer support method, self-support method, and substrate method. The most commonly used method is gelation, in which graphene sheets are physically or chemically connected to each other with cross-linking agents to form a space network.[52, 53] In the process of template method, organic molecules or their self-assembly system are usually used as template, and the graphene sheets are guided by the template agent through the effect of hydrogen bond, ionic bond and wan der Waals force. Thus, the resulting graphene gel is structurally controllable and highly ordered .[54–56] In the spacer support method, gaskets are added into the graphene layers to prevent the invalid reunion of graphene sheets.[57, 58] Opposed to a spacer support method, the self-support method refers to form the graphene gel using alkali activation or gas isolation effect, without template or gasket. Substrate method is mainly used to form thin graphene film,[59] chemically modified graphene solution is dropped into horizontal substrate and dried.[60–62]

CNTs are exciting candidates for electrically conducting aerogels. They can form electrically percolating networks at very low volume fractions and elastic gels in concentrated suspensions through van der Waals interaction mediated cross-linking.[45] CNTs can directly cross-link with each other to obtain aerogels in the collection process. However, such aerogels always have poor 3D space structure with bad mechanical strength and elasticity. To eliminate these drawbacks, the gelation stage is introduced. Meanwhile, a certain amount of surfactant is used to enhance the surface activity of carbon nanotubes, and polymer is added to reinforce the networks.[45, 46, 63,64]

Natural biomass is receiving increasing attention to prepare carbon nanostructures due to their low cost, rich source, non-toxicity to humans, etc. Cellulose rich biomasses are usually chosen as precursor for the preparation of aerogels. For the hydrated substance, like the winter melon [65] and bacterial cellulose,[21, 66, 67] the gelation stage is not necessary. The 3D web structure of hydrated substance contains a large void volume and absorbs a high portion of water (over 80wt.-%), after water sublimation during freeze drying, the porous framework of the 3D aerogels is obtained. However, the substance with low water content usually needs to be homogenized to form a suspension, following with the gelation and freeze drying.[7,68]

2.2 Drying

Drying is the critical stage for carbon aerogel preparation. In this stage, the solvent consisted in the hydrogel is removed leaving only solid network, i.e aerogel. The drying process significantly affects the final structure and properties of aerogels. Three drying methods are commonly employed (Figure 3): (1) supercritical drying,[1, 69] the elimination of the solvent occurs under supercritical conditions; (2) freeze drying,[16,17,70] the solvent is frozen and removed by sublimation; and (3) ambient drying,[5, 41, 71] the solvent is dried by simple evaporation under atmospheric environment. However, in the ambient pressure drying process, the capillary tension at solid–liquid–vapor interface could cause the structural collapse and dimensional shrinkage of hydrogel. Therefore, the most commonly used processes at present are supercritical drying and freeze drying.[72]

2.2.1 Supercritical Drying

Supercritical drying is the most appropriate and efficient process to dry the hydrogels.[73] Theoretically, during the supercritical drying process, the pressure and temperature of vessel overpasses the critical point (TC, PC) of the solvent, and there are no discriminations in liquid or vapor phase. Therefore, the surface tension at the solid-liquid-gas interface is eliminated and the shrinkage and collapse of the pores are avoided. Compared with other drying methods, the obtained aerogels have better performance, such as higher porosity and pore volume. However, the gel is slightly shrunken in practical application due to the incomplete solvent replacement. Simultaneously, the smaller the pore size, the more the pore shrinkage; and the lower the precursor concentration, the more the structure collapse.[74] The drying effect is also related to the type of solvent, and CO_2 is commonly employed because of the milder critical pressure and temperature.[75, 76] However, there are some disadvantages for supercritical drying, such as high pressure required (Pc=7.4 MPa) and long solvent replacement time spent. In conclusion, supercritical drying is one of the best ways to keep the initial pore structure of gel and reduce shrinkage and collapse. But it is difficult to achieve large-scale production and commercial applications due to the high cost.[77]

2.2.2 Freeze Drying

Freeze drying is an environmentally friendly drying process.[78] Like supercritical drying, the pore structure can be maintained in freeze drying by eliminating the surface tension.[79] During this drying process, the solvent in the gels is frozen and eliminated by sublimation at low pressure, avoiding the formation of a gas-liquid interface. The properties of the resulting aerogels are influenced by lots of factors, such as pre-drying treatments, freezing rate, pore size, porosity, precursor concentration, etc. A long aging period can improve the mechanical strength of the gel. Also, a solvent with low expansion coefficient and high sublimation pressure can reduce the structure shrinkage and collapse, and the most commonly employed solvent is water. Freezing rate can affect the pore structure by changing the shape and size of ice crystals, a rapid freezing rate leads to the formation of small ice crystal, results in a small pore size and high specific surface area.[80] When the pore size of the gel is too small, the water in the hole is difficult to freeze, resulting in the shrinkage and collapse of the aerogels. Therefore, freeze drying is not suitable for the preparation of carbon aerogels with small pore size. In addition, a low precursor concentration results in the growth of ice crystals, which will greatly change the pore structure of aerogels. Hence, as a replacement of supercritical drying, freeze-drying is mainly applied to prepare aerogels with high requirements on macroscopic shape.[81]

2.2.3 Ambient Drying

Ambient drying is a simple and safe technique with low cost, so it is the most promising way for the large-scale preparation of aerogels in industrial production.[82] However, in this method, the capillary tension at solid–liquid–vapor interface could cause the structural collapse and dimensional shrinkage. The larger the pore size, the smaller the liquid surface tension; the larger the interface contact angle, the smaller the capillary pressure of gas-liquid interface. There are two approaches to prepare aerogels with low density and good pore structure under ambient drying: (1) reducing the solvent surface tension to reduce capillary tension; (2) strengthening the framework structure to resist capillary tension. Methods to reduce solvent surface tension mainly include solvent replacement [83] and surfactant addition.[84] When the strength of the gel framework is enough to resist the capillary pressure, the collapse and shrinkage of the pore structure can be avoided. As the R/C ratio increased from 50 to 1000 (the concentration of alkali catalyst decreased continuously), the shrinkage rate of aerogels decreased from 68 % to 33 %. In other words, the shrinkage during liquid evaporation is reduced with the increasing of gel junction. Therefore, the physical properties of the resulting aerogels are similar to those obtained through supercritical drying when the reactant concentration is increased or the catalyst concentration is decreased.

2.3 Carbonization

In the carbonization process, the aerogel is heated to a high temperature (≥600 °C) under the inert atmosphere of N_2 or Ar, oxygen and hydrogen moieties are decomposed into gases through high temperature reaction and escaped from the aerogel to form carbon structure.[85] Notably, the stage is significant for polymer carbon aerogels and biomass carbon aerogels, but it's unnecessary for those ones based on graphene and CNTs due to the completely graphitized structure. The physical properties of aerogels are significantly changed in this stage. The macropore in aerogel is decreased due to the shrinkage (ca. 20 %) of the structure, while the mesopore and micropore are increased, especially the micropore. The micropore performance of carbon aerogels is mainly determined by the carbonization process, while the mesoporous and macroporous properties are mainly determined by the initial preparation parameters (such as the concentration of precursor and catalyst) and the drying conditions.[86, 87] Additionally, the specific surface area is significantly increased by the carbonization treatment, especially at the lower temperature. The possible reason is that the oxygen and hydrogen groups escaped from the framework, leaving lots of new holes (mainly micropores). The aerogels exhibit a part of graphite characteristics after carbonization, but this is only when the temperature exceeds 2000 °C that the complete graphitization structure appears.

The carbonization temperature is one of the most important parameters to determine the physical properties of carbon aerogels. Temperature changing do not significantly influence the specific surface area of aerogel when it is below 600 °C. While the pore volume decreases with the increase of temperature when it is in the range of 700 °C to 900 °C. However, the pore volume is not change obviously with the temperature increased beyond 900 °C.[50, 86] In addition, the yield is not affected by the changing of carbonization temperature when it is above 750 °C, remaining at around 50 %.

3. Applications Based on Adsorption Performance

To preserve the environmental quality, control and remove of hazardous organic compounds (oils, organic solvents, VOCs, etc.), heavy metal ions, nitrogen oxides (NO_x; NO, NO_2, N_2O, etc.), carbon monoxide (CO), greenhouse gases i.e. methane (CH_4) and carbon dioxide (CO_2), etc. are strictly necessary. The characteristics of large specific surface area, high porosity and adjustable surface chemistry endow carbon aerogels with excellent adsorption capacity. Therefore, they are widely used as attractive adsorbents. In environmental protection field, carbon aerogels have a good adsorption capacity for most of the mainstream pollutants.

3.1 Applications for Water Treatment

3.1.1 Oil/Water Separation

In recent decades, accidents of oil leakage and oil spill have become one of the top environmental concerns with the fast-growing exploitation of crude oil.[88, 89] Selective oil/water separation is highly desired to reclaim the precious oil resources. Therefore, the development of selective, efficient and eco-friendly oil sorbents for oil/water separation is extremely urgent. The selective oil absorption capacity is mainly determined by two properties of the absorbent. The first is surface hydrophobicity, the higher surface hydrophobicity, the less competing adsorption of water. At the same time, the high surface

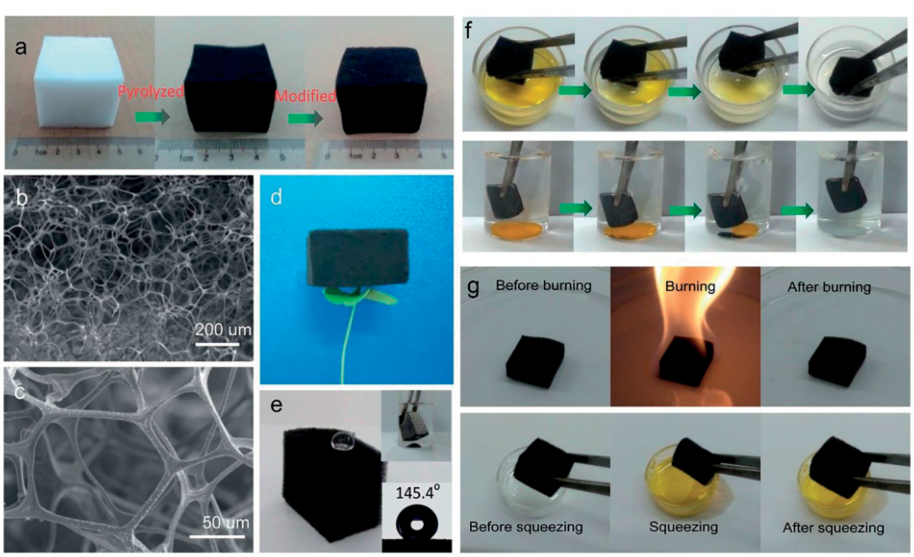

Figure 4. (a) Fabrication of the carbon aerogel. (b and c) SEM images of the carbon aerogel at different magnifications. (d) A 15 cm^3 carbon aerogel standing on a piece of *Marsilea quadrifolia* grass. (e) Hydrophobicity of the carbon aerogel. (f) Adsorption of organic liquids by the carbon aerogel. (g) The regeneration of the carbon aerogel. Adapted with permission from ref.[93] Copyright (2014) The Royal Society of Chemistry.

hydrophobicity lead to the spontaneous oil uptake. The surface hydrophobicity can be improved by adjusting the surface functional groups and rationally designing the hierarchical surface textures. The second is the density of carbon aerogels. The density is determined by the porosity, and the pores provide the space for storage of the absorbed oils. Therefore, the density is inversely correlated to the calculated absorption capacity. In a word, an applicable absorbent for selective oil/water separation should have the following characteristics: low density and high porosity, low surface energy and high hydrophobicity, robust mechanical strength, good recyclability, and scalable preparation.

Carbon aerogels are frequently used as the selective adsorbent for oil/water separation due to the intrinsic characteristics.[18,64,90–95] Oxygen and hydrogen containing groups in aerogels are decomposed in the hydrothermal or carbonization process, leaving the graphitization structure, which is mainly composed by low-surface-energy sp^2 carbon. Wang et al.[92, 93] fabricated nitrogen-rich carbon aerogels using poly foams (melamine formaldehyde) as precursors. The resulting aerogel possessed an amazing capacity for selective adsorption of oils from water up to 158 times its own weight. Meanwhile, the aerogel could be easily recycled due to the fire resistance and compressibility properties (Figure 4). Chen et al. [94] prepared a new carbon aerogel by the direct carbonization of melamine foam. The aerogel exhibited high porosity (over 99.6 %), low density (5 mg cm^{-3}), high specific surface area (268 $m^2 g^{-1}$), and excellent absorptive properties towards oil and organic solvents (148 to 411 times its own weight).

Graphene aerogels, which are fabricated with graphene oxide (GO) by self-assembly and reduction, have proven their capacity as super absorbents for oil/water separation.[58,96–106] For example, Xu et al. [97] fabricated graphene aerogel by hydrothermal chemical reduction and freeze-drying processes. The oxygenic groups were reduced by L-phenylalanine during the hydrothermal process, resulting excellent superhydrophobicity and superoleophilicity. When used as adsorbent for selective oil/water separation, the adsorption capacity was higher than 100 times its own weight. Carbon nanotubes are emerging as a simple 1D material with excellent hydrophobicity and high surface area, and the CNTs aerogels are also familiarized with people as a selective adsorbent for organic pollutants.[27,107–109] For example, Gui et al. [107] reported a magnetic CNTs aerogel with a high mass sorption capacity for diesel oil with 56 g g^{-1}. And the aerogel can be recycled by magnetic force and desorption by simple heat treatment. Moreover, CNTs were added to the graphene aerogel to improve the mechanical strength and specific surface area. The composite aerogels are more suitable for environmental clean-up, especially for oil/water separation.[18,24,27,109–112] Zhan et al.[18] creatively introduced PDA-functionalized carbon nanotubes into graphene aerogel to form a robust PDA/MWCNT/graphene composite aerogel with a novel "cabbagelike" hierarchical porous structure (Figure 5). The miniaturization of pore size around the composite aerogel periphery enhanced the capillary flow in aerogel channels, and the super-absorption capacity for organic solvents was up to 501 times to its own mass. Besides, the composite aerogel exhibited excellent reusable performance in absorption–squeezing, absorption–combustion, and absorption–distillation cycles.

Although polymer-, graphene- and CNT- based aerogels exhibited fascinating capacity for the selective oil/water separation, they are not suitable for application in the industry due to their high cost and complex preparation process. Recently, biomass materials (even waste carbonaceous materials) have been converted into hydrophobic carbon aerogels and used for oil/water separation.[7,64,90,113–118] Li et al. [91] exploited a carbon aerogel by pyrolysis of poplar catkin microfibers with tubular structure. The resulting carbon aerogel showed excellent performance, like ultralow density, high compressibility, high electrical conductivity. The absorption capacity for oil was up to

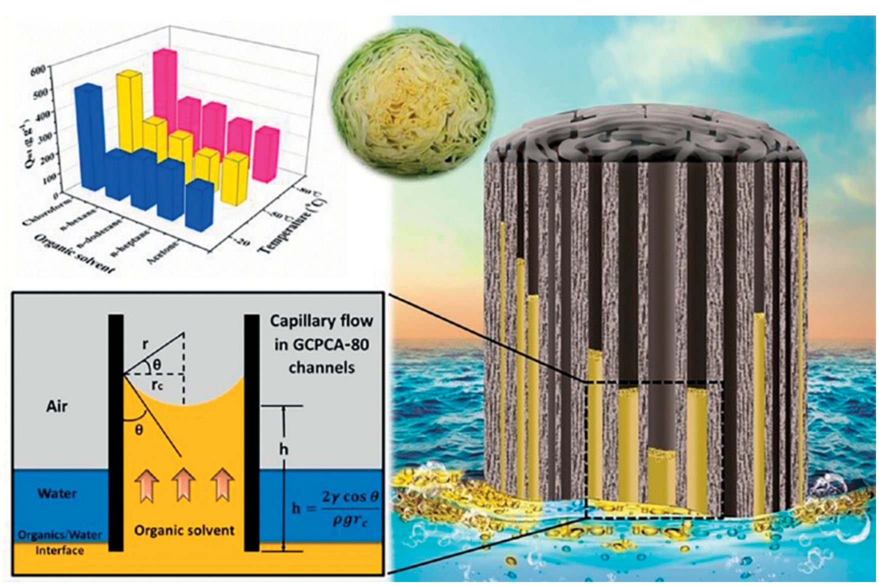

Figure 5. The absorption mechanism model of the directional alignment capillary flow in PDA/MWCNT/graphene composite aerogel with "cabbagelike" hierarchical porous channels. The top left picture shows the super-absorption capacity of the composite aerogel frozen at different temperatures for organic solvents. Reprinted with permission from ref.[18] Copyright (2018) American Chemical Society.

Table 1. Summary of carbon aerogels and their performance in selective oil/water separation.

Kinds of carbon aerogels	Absorbed substances	Sorption capacity	Refs.
Graphene aerogels	Carbon tetrachloride, Toluene, Chloroform, Petroleum ether, Acetone, DMF, Dichloromethane, Ethanol, *n*-Hexane, Pump oil, Paraffin liquid	70.0–195.0 g g^{-1}	[17]
PDA/CNTs/graphene aerogels	Chloroform	501.0 g g^{-1}	[18]
CNTs/graphene aerogels	Crude oil, Peanut oil, *n*-Hexane, Diesel oil, Octane, Gasoline, Hexadecane, Engine oil, Chloroform, PDMS, ODCB	100.0–322.8 g g^{-1}	[24]
Graphene/CNTs aerogels	*n*-Hexane, Toluene, Phenixin	140.0–200.0 g g^{-1}	[27]
Poplars catkin carbon aerogels	Chloroform, Dichloromethane, Soy bean oil, Pump oil, Petroleum ether, Toluene, *n*-Hexane	80.0 – 161.0 g g^{-1}	[91]
Lignin/graphene aerogels	Petroleum oils, Crude oil, *n*-hexane, Chloroform, Methanol, Ethanol, Decane, Toluene, Motor oil, Carbon tetrachloride, Corn oil	254.0–522.0 g g^{-1}	[96]
Polydopamine/chitosan/graphene aerogel	Ethanol, Gasoline, Acetone, Diesel, Hexane, Toluene	10.0–21.0 g g^{-1}	[98]
Graphene/polyurethane aerogel	*n*-Hexane, Diesel oil, Lubricating oil, Crude oil	25.8–44.1 g g^{-1}	[99]
PVA/CNTs aerogels	DMF, Chloroform, THF, Acetonitrile, Acetone, Hexane, Toluene, Bean oil	10.0–52.0 g g^{-1}	[109]
Lignin/agarose/PVA carbon aerogels	Tetrahydrofuran, Toluene, Oleic acid, Dichloromethane, Chloroform, Hexane, Vacuum pump oil, Liquid paraffin	20.0 – 40.0 g g^{-1}	[118]
Graphene/nanofiber aerogels	Phenixin, Chloroform, Dichloroethane, Dioxane, Olive oil, Pump oil, Toluene, *p*-xylene, Ethanol, *n*-hexane	230.0–734.0 g g^{-1}	[119]
Cellulose/graphene aerogels	Benzene, Decane, Dichloromethane, Xylene, DMSO, Olive oil, Gasoline, Octane, Toluene, Hexane, Chloroform	80.0–197.0 g g^{-1}	[120]

161 times its own mass. Jiang et al.[118] prepared a biomass-derived "honeycomblike" aerogel with lignin, agarose, and poly (vinyl alcohol). The aerogel was then immersed into the solution containing pH responsive component poly (2-(dimethylamino) ethyl methacrylate). The finally modified aerogel could switch the adsorption to desorption by controlling the medium pH. The modified aerogel was also featured high oil adsorption capacity with 40 times its own weight. The recent researches of selective oil/water separation by carbon aerogels were summarized in Table 1**.**

3.1.2 Removal of Heavy Metal Ions

Heavy metal ions, including chromium (Cr), copper (Cu), plumbum (Pb), mercury (Hg), manganese (Mn), cadmium (Cd), nickel (Ni), zinc (Zn) and iron (Fe), etc. are also one of the most important pollutants in the aquatic environment. And they are mainly discharged from the industrial emissions and effluents. Heavy metals are known for the high toxicity and carcinogenicity to human beings, and they can residual in the eco-environment for a long term due to the nonbiodegradability.[121] Therefore, removal of heavy metal ions from the contaminated environment is highly desired. Among the different treatment processes, adsorption is the most suitable approach with many advantages.[122] For heavy metal removal, the mechanisms are mainly based on the physical adsorption, and the adsorption capacity is seriously influenced by surface affinity groups of adsorbents. Different from the oil/water separation, the more hydrophilic groups containing N, O, S, P on the surface of adsorbents, the stronger adsorption capacity of heavy metals.

Carbon aerogels are widely used in heavy metal adsorption because of their porous structure, large specific surface area, high porosity and, most importantly, controllable surface functional groups.[31,116,123–130] In earlier times, Meena et al. [123] investigated the adsorption of seven kinds of heavy metals (Cd, Pb, Hg, Cu, Ni, Mn and Zn) using carbon aerogel (carbonized from resorcinol formaldehyde (RF) aerogel). The adsorption capacity has been found to be pH, temperature, adsorbent dose, concentration and contact time dependent, and the pH was the most effective parameter. It has been demonstrated that the ion exchange process and complexation were major mechanisms. The highest adsorption capacity was obtained for Cd (II) (400.8 mg g^{-1}) while lowest capacity for Pb (II) (0.70 mg g^{-1}). The disparity was caused by the different chemical affinity and ion exchange capacity of different heavy metal ions toward the sorbent functional groups. In addition, the Langmuir model could be used to explain the adsorption behavior of all metal ions, illustrating that the molecules formed a monolayer on the substrate surface with minimum interactions, and the adsorption energy was equal for all sites.

In the following years, scientists focused on the study of how to improve the adsorption performance of carbon aerogels on heavy metal ions, including increase the porosity and specific surface area, change the surface composition and expose the active sites and so on. For example, Petra et al. [125] prepared N-doped carbon aerogels by ammonia assisted pyrolysis, and the aerogel was used for removing Cu (II) and Pb (II) ions. They found that the specific surface area, micropores surface area and micropores volume were increased in NH_3 pyrolysis process, and the oxygen content of the N-doped samples was higher than that of non-doped. Moreover, higher hydrophilicity was revealed for N-doped carbon aerogel, and the adsorption capacity was improved 2–6 times in comparison with the non-doped samples. Similarly, Alatalo et al. [117] synthesized a N-doped carbon aerogel using globular protein as the natural source of nitrogen functionalities and structural directing agent, the aerogel was applied to remove Cr (VI) and Pb (II) ion in wastewater. The N-doped carbon aerogel showed a super ability to adsorb both Cr (VI) (ca. 68.0 mg g^{-1}) and Pb (II) (ca. 240.0 mg g^{-1}), because of the acidic character enhancing Pb (II) removal through electrostatic interaction and Cr (VI) removal by reduction. Authors also found that the removal of Pb (II) was controlled by chemical reactivity and Cr (VI) removal was controlled by pore diffusion and chemical reaction on adsorbent surface. Furthermore, the presence of competing ions showed little effect on the adsorption efficiency toward Cr (VI) and Pb (II).

Heavy metal adsorption capacity of carbon aerogels can also be enhanced through the adjustment and control of structure and the enlargement of exposed active sites. Chen et al. [110] constructed a 3D aerogel through self-assembly of 2D graphene oxide nanosheets and 1D carbon nanotubes for removal of heavy metal ions. The adsorption capacity of 3D aerogel (Cd^{2+}, 235 mg g^{-1}) was higher than other adsorbents due to the synergistic effects of GO and CNTs in micro-environment, nano-substrate, and active sites. Firstly, GO sheets were highly separated into monolayer form and created large interspaces. Secondly, GO and CNTs provided abundant adsorption sites and a great amount of them were exposed. Thirdly, the adjustment of the interspaces micro-environment resulted in the selective adsorption capacity. Moreover, the aerogel exhibited superior adsorption capabilities to organic pollutants (oxytetracycline, 1729 mg g^{-1}, diethyl phthalate, 680 mg g^{-1}, methylene blue, 685 mg g^{-1}, and diesel, 421 mg g^{-1}) as well as heavy metal ions (Cd^{2+}, 235 mg g^{-1}). These have shed new light on the simultaneous removal of organic pollutants and heavy metal ions in the combined wastewater. Kong et al. [128] synthesized a nitrogen (N) and sulfur (S) co-doped graphene-based aerogel for simultaneous adsorption of Cd^{2+} and organic dyes in printing and dyeing wastewater. The adsorption capacity towards the Cd^{2+} was enhanced by the presence of the dyes, while the capacities towards the dyes were not affected by the presence of Cd^{2+}. At the same time, they found that the surface and intraparticle diffusion was the key factor in controlling the rate of adsorption.

Recently, lots of low-cost biomass-based carbon aerogels were applied for heavy metal ion removal. Chen et al. [130] developed a biomass carbon aerogel with cotton using alkaline etching methods, and used to remove Co (II), Cd (II), Pb (II) and Sr (II) from aqueous solutions. The adsorption capacity of cotton aerogels for heavy metal ions were much higher than that of raw cottons. This phenomenon demonstrated that the adsorption capacity of heavy metal ion was strongly influenced by the interaction between these ions with the oxygen containing functional groups on the aerogel surface. There are many other researches about the removal of heavy metal ions by carbon aerogels, and the adsorption capacities are summarized in Table 2.

Basically, adsorbent regeneration is an important step of the removal of heavy metal ions. Up to now, the using of acids, alkalis, salts, deionized water, chelating agents and buffer solutions have been reported.[131] However, there is less research implemented for the regeneration of saturated carbon aerogels, and the desorption mechanism of heavy metal ions is still unknown. Therefore, more efforts are needed to explain the desorption of heavy metal ions and the regeneration of carbon aerogels.

3.2 Applications in Air Cleaning

3.2.1 Removal of Volatile Organic Compounds (VOCs)

Volatile organic compounds (VOCs) are a big family of carbon-based chemicals that participates in atmospheric photochemical reactions, excluding CO_2, CO, carbonic acid, metallic carbides or carbonates, and ammonium carbonate. There are more than 300 types of VOCs, mainly including alcohols, aldehydes, alkenes, aromatic compounds, halogenated VOCs, ketones, and polycyclic aromatic hydrocarbons. All of them have the same characters of low boiling point, high vapor pressure, and strong reactivity, and they are a major kind of pollutants in the air. Most of the VOCs are harmful for both the human health and ecological environment. As for the human health, they are

Table 2. Summary of carbon aerogels and their performance in heavy metal ion capture.

Kinds of carbon aerogels	Absorbed substances	Sorption capacity	Refs.
Graphene oxide aerogels	Cu (II)	19.13 mg g^{-1}	[31]
Cellulose carbon aerogels	Cr (VI)	68.15 mg g^{-1}	[117]
	Pb (II)	40.00 mg g^{-1}	
Resorcinol formaldehyde polymerized carbon aerogels	Cd (II)	400.8 mg g^{-1}	[123]
	Cu (II)	561.7 mg g^{-1}	
	Hg (II)	45.62 mg g^{-1}	
	Ni (II)	12.91 mg g^{-1}	
	Zn (II)	1.84 mg g^{-1}	
	Mn (II)	1.32 mg g^{-1}	
	Pb (II)	0.71 mg g^{-1}	
Graphene carbon sphere aerogel	Cr (VI)	12.69 mg g^{-1}	[124]
	U (VI)	13.31 mg g^{-1}	
N-doped carbon aerogels	Cu (II)	19.06 mg g^{-1}	[125]
	Pb (II)	51.75 mg g^{-1}	
Nanocellulose aerogels	Hg (II)	718.5 mg g^{-1}	[126]
Graphene/nanotubes carbon aerogels	Cd (II)	235.0 mg g^{-1}	[127]
N, S-doped graphene aerogels	Cd (II)	197.3 mg g^{-1}	[128]
Cotton carbon aerogels	Co (II)	71.43 mg g^{-1}	[130]
	Cd (II)	40.22 mg g^{-1}	
	Pb (II)	111.1 mg g^{-1}	
	Sr (II)	33.32 mg g^{-1}	
Waste paper carbon aerogels	Cu (II)	156.3 mg g^{-1}	[132]

highly toxic and carcinogenic, and they cause respiratory issues, acute poisoning or chronic toxicity as well as nasal tumors. And for the ecological environment, they bring serious problems including greenhouse effect, photochemical smog, stratospheric ozone depletion etc. Therefore, selective removal of VOCs is highly desired to air clean-up. Recently, tremendous efforts have been made to develop efficient VOCs elimination techniques. Adsorption process has been considered one of the most efficient methods with low cost and no secondary organic pollutants. And the key point of adsorption process is developing the high-performance adsorbents. Same as oil absorption, the VOCs absorption capacity is determined by surface hydrophobicity and the density of adsorbents.

Carbon aerogels have been recognized as potential adsorbents for VOC removal due to their superior features.[90,133–147] For example, Francisco et al. [136] prepared a carbon aerogel with gallic acid, resorcinol, and formaldehyde, which was further activated by CO_2. The resulting carbon aerogel exhibited a fascinating pore texture constituted by a monomodal mesoporous network superimposed over a microporous network. The adsorption capacity towards VOCs was tested by the dynamic adsorption of benzene, toluene, and xylenes in dry and wet air flow. They found that the adsorption capacity in dry air was determined by the total microporosity, and it was increased in the order benzene < toluene < xylenes. Though the similar order was obtained with wet air, the adsorption amounts were slightly lower than that from the dry air due to the competition of water. Han et al. [144] constructed a 3D hierarchical porous graphene aerogel for efficient adsorption and preconcentration of chemical warfare agents. The resultant aerogel exhibited a higher adsorption capacity towards VOCs at comparatively high relative humidity than that of ordinary graphene aerogel. Authors demonstrated that the capacity was given by the facilitating of contaminated gas diffusion of 3D hierarchical porous structure and the enlarging of the interaction between VOCs molecules and the active adsorption sites through hydrophobic interaction. The recent researches of VOC removal by carbon aerogels were summarized in Table 3.

Another important application of carbon aerogels for VOC adsorption is their use as VOC sensors .[142, 143, 145, 147] The adsorption of trace amounts of gas molecules on the carbon aerogel surface causes significant charge transfer, resulting in a noticeable conductance change. Dolai et al. [142] constructed a new hybrid PDA–aerogel by embedding the PDA within aerogel pores. The PDA–aerogel composite exhibited rapid color/fluorescence response and enhanced signals upon exposure to VOCs at low concentrations (Figure 6). The encapsulated PDA derivatives formed a PDA–aerogel “artificial nose” through the production of distinct VOC-dependent color transformations. However, simultaneous detection of different kinds of trace VOCs with high sensitivity and selectivity is still challenging.

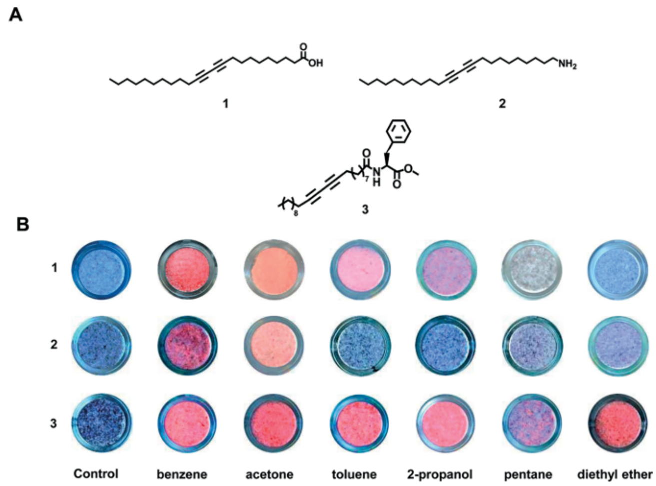

Figure 6. “Color fingerprinting” of VOCs using aerogel-embedded PDA derivatives. (A) Structures of the monomers employed; (B) scanned photographs of the PDA-aerogels after 60 min exposure to the VOCs (concentrations 1000 ppm). Reprinted with permission from ref.[142] Copyright (2017) American Chemical Society.

Wu et al. [145] reported a chemiresistor-type sensor with 3D sulfonated reduced graphene oxide aerogel and used it for VOC detection. They found that the responses of $NaHSO_3$ functionalized sensor to methanol, ethanol, acetone, toluene, and chloroform saturated vapors reached as high as 40 %, 33 %, 30 %, 25 %, and 16.2 %, about two orders of magnitude are higher than that of unfunctionalized sensor. At the same time, the sensor exhibited a counter response toward to an electron-withdrawing gas, and a positive response to electron-donating saturated VOCs (CH_3OH, C_2H_5OH, and $CHCl_3$ etc.). And the characteristic patterns of response–temperature curves were employed to distinguish various gases.

The adsorption capacity of carbon aerogels to VOCs can be influenced by many factors, basically from adsorbents (specific surface area, pore size, and surface functional groups), adsorbates (molecular structure, molecular polarity and boiling point), and adsorption conditions (temperature, humidity, and flow velocity). Lots of work needs to be done to enhance the VOC removal rate by carbon aerogels, like improving adsorption

Table 3. Summary of carbon aerogels and their performance in removal of VOCs.

Kinds of carbon aerogels	Absorbed substances	Sorption capacity	Refs.
Activated carbon aerogels	Methanol, Acetone, Cyclohexane, Benzene	1435–1750 mg g^{-1}	[133]
Resorcinol/formaldehyde carbon aerogels	Benzene, Toluene, Xylenes	307.0–380.0 mg g^{-1}	[134]
Resorcinol/formaldehyde carbon aerogels	Toluene	1180 mg g^{-1}	[135]
Gallic acid–resorcinol carbon aerogels	Benzene, Toluene, Xylenes	360.0–450.0 mg g^{-1}	[136]
Carbon–silica aerogels	Benzene	395.2 mg g^{-1}	[138]
Resorcinol/formaldehyde carbon aerogels	Toluene, Acetone	174.0–715.0 mg g^{-1}	[139]
Resorcinol/formaldehyde carbon aerogels	*n*-Hexane	150.0–240.5 mg g^{-1}	[141]
Graphene aerogels	Dimethyl methylphosphonate, Sarin	38.4–148.0 mg g^{-1}	[144]

capacity, decreasing adsorbent production costs, increasing the adsorption selectivity, and so on.

3.2.2 CO_2 Capture

As a crucial greenhouse gas, CO_2 clearly contributes to the global warming. In the 21st century, the increase of CO_2 concentration become one of the most serious problems for human beings.[148] Scientists have predicted that the concentration of CO_2 would exceed 550 ppm in 2050, and it will lead to a series of diseases for people if the concentration is above 600 ppm.[149] Therefore, reducing CO_2 emissions has never been more urgent than nowadays. Among different CO_2 capture methods, physical adsorption by porous materials is identified as an attractive option. Different CO_2 sorbents may vary with application, but all sorbents must have the following characteristics: (i) High CO_2 adsorption capacity; (ii) High selectivity for CO_2 adsorption; (iii) Mild conditions for regeneration; (iv) High stability and resistance; (v) Fast adsorption kinetics.

As a three-dimensional hierarchical porous material, carbon aerogels have a high total pore volume and specific surface area, resulting the enhanced diffusion and adsorption of CO_2 molecules. In addition, the microstructure, composition, and surface chemistry of carbon aerogels can be purposefully manipulated due to the multifunction of the sol–gel process. Therefore, carbon aerogels are widely used in CO_2 capture.[33,150–165] Generally, the CO_2 adsorption capacity of carbon aerogels is improved with the increase of pores and pore interconnection. Robertson et al. [150] synthesized carbon aerogels through sol-gel process and subsequently activated with KOH, obtained a high pore volume of 2.03 cm^3 g^{-1} and a high surface area of 1980 m^2 g^{-1}. The proportion of microporosity was as high as 87 %, and the micropore size distribution centered at ca. 8 and 13 Å. The CO_2 capture capacity of the carbon aerogel was as high as 2.7–3.0 mmol g^{-1}. In another study, Dassanayake et al. [163] prepared activated carbon aerogels with chitin by pyrolysis and KOH activation. The results show that the specific surface area and volume of micropores were increased by about 37-fold and 95-fold by KOH activation. This resulted in a high CO_2 adsorption capacity of 5.02 mmol g^{-1} at 1.0 atm, 0 °C and 3.44 mmol g^{-1} at 1.0 atm, 25 °C. Alhwaige et al. [33] firstly synthesized carbon aerogels using chitosan and polybenzoxazine as precursors, and the aerogel was reinforced by montmorillonite. The as-prepared carbon aerogels had a large pore size and high BET surface area and exhibited an excellent CO_2 uptake capacity with a maximum of 5.72 mmol g^{-1}.

An ideal CO_2 adsorbent should simultaneously have the characteristics of large adsorption capacity, adsorption selectivity as well as desorption capacity. Therefore, the adsorption energy of adsorbents for CO_2 capture should be appropriate to achieve the characteristics. And the adsorption energy can be controlled by adjust the surface chemical properties. Youngtak et al. [157] constructed a reduced graphene oxide aerogel and functionalized with carbon nitride. The functionalized aerogel exhibited a large CO_2 capture capacity of 0.43 mmol g^{-1} at 1.0 atm, 25 °C, and a high CO_2 selectivity against N_2. Moreover, 98 % CO_2 was desorbed easily by simple pressure swing. DFT analysis found that the CO_2 adsorption capacity, selectivity and regenerability were increased through induced dipole interaction by microporous edges of graphitic carbon nitride (Figure 7).

Amine functionalized carbon aerogels are a promising material for CO_2 capture due to the strong affinity of amine moiety to CO_2 molecules. In the presence of moisture, amine moiety reacts with CO_2 selectively ammonium carbamate, while the product is carbamic acid in the dry atmosphere. The reaction is reversible and the saturated carbon aerogels is regenerated completely just by heating up to 100 °C. Therefore, more and

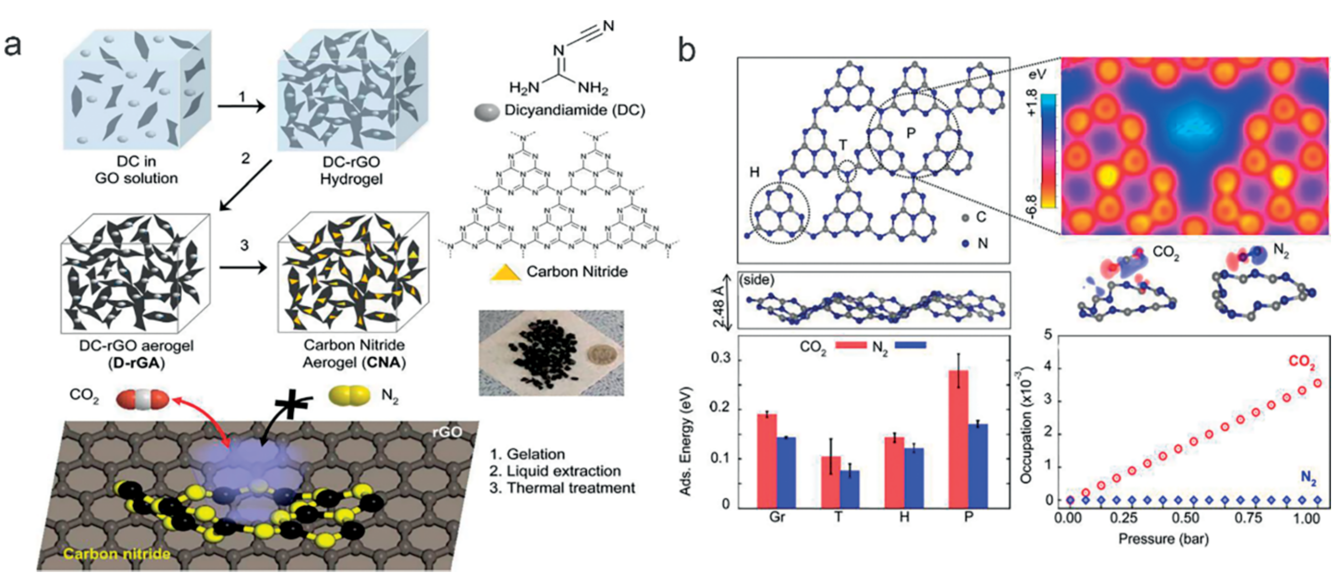

Figure 7. (a) Synthetic procedure for carbon nitride aerogel (CNA). Modified sol-gel reduction of graphene oxide (GO) with dicyandiamide results in dicyandiamide-functionalized reduced graphene oxide (DC-rGO) hydrogel. Subsequent liquid extraction with supercritical CO_2 drying and thermal treatment transform this hydrogel into carbon nitride embedded graphene aerogel, which is capable of selective CO_2 gas capture against N_2. (b) DFT analysis of gas adsorption by carbon nitride aerogel. Reprinted with permission from ref. [157] Copyright (2015) American Chemical Society.

Table 4. Summary of carbon aerogels and their performance in CO_2 capture.

Kinds of carbon aerogels	Surface modifications	Sorption capacity	Adsorption condition	Refs.
Chitosan polybenzoxazine aerogels	Montmorillonite	5.72 mmol g^{-1}	1.0 atm, 25 °C	[33]
Resorcinol/formaldehyde carbon aerogels	Activated by KOH	3.00 mmol g^{-1}	1.0 bar, 25 °C	[150]
Graphene oxide aerogels	Polyethylenimine	2.55 mmol g^{-1}	1.0 bar, 0 °C	[151]
Resorcinol/melamine aerogels	–	2.70 mmol g^{-1}	1.0 bar, 25 °C	[153]
Resorcinol/formaldehyde/silsesquioxane aerogels	Amine	2.57 mmol g^{-1}	1.0 atm, 25 °C	[155]
Activated carbon aerogels	Activated by KOH	4.90 mmol g^{-1}	1.0 bar, 0 °C	[156]
Graphene oxide aerogels	Carbon nitride	0.43 mmol g^{-1}	1.0 atm, 25 °C	[157]
Aldehydes/amines aerogels	–	14.10 mmol g^{-1}	1.0 bar, –78 °C	[158]
		1.50 mmol g^{-1}	1.0 bar, 25 °C	
Cellulose carbon aerogels	N-doped	4.99 mmol g^{-1}	1.0 atm, 25 °C	[159]
Cellulose carbon aerogels	Activated by CO_2	3.42 mmol g^{-1}	1.0 atm, 25 °C	[161]
Resorcinol/formaldehyde aerogels	–	13.00 mmol g^{-1}	35.0 bar, 35 °C	[162]
Chitin carbon aerogels	Activated by KOH	5.02 mmol g^{-1}	1.0 atm, 0 °C	[163]
		3.44 mmol g^{-1}	1.0 atm, 25 °C	
Cellulose nanofibrils aerogel	Amine	1.91 mmol g^{-1}	1.0 bar, 25 °C,	[164]

more amine moieties, including polyethyleneimine, alkanolamines, diethanolamine, monoethanolamine, and methyldiethanolamine, were used to modify the carbon aerogels to increase the selectivity towards CO_2.[151, 155, 160, 164] Sui et al.[151] fabricated a 3D porous aerogel with graphene oxide (GO) sheets and polyethylenimine (PEI). The pore-rich and amine-rich structure of the aerogel resulted in a higher adsorption capacity for carbon dioxide (11.2 wt.-% at 1.0 bar and 273 K) than that for GO (7.5 wt.-% at 1.0 bar and 273 K). Kong et al.[155] constructed an amine hybrid resorcinol–formaldehyde/silica composite aerogel by a sol-gel process combined with supercritical drying. The results of CO_2 capture tests revealed that the amine existence improved the CO_2 adsorption capacities. The same function was found with the existence water, especially in improving the useful adsorption capacity. The total CO_2 adsorption capacity at 25 °C was as high as 2.57 mmol g^{-1}, while the useful CO_2 adsorption capacity was 1.85 mmol g^{-1}. Different aerogel-based adsorbents and their CO_2 capture performance are summarized in Table 4.

4. Applications Based on Catalysis

Catalysts play an essential role in treating environmental pollutants on the road to a sustainable and clean environment. The extraordinary properties, particularly electrical conductivity, help carbon aerogels good at electron transfer. Therefore, carbon aerogels exhibit an attractive catalytic performance for the chemical degradation of the hazardous compounds. In this field, carbon aerogels are mainly exploited as an attractive catalyst or catalyst supports.[13,166]

4.1 Use as Catalyst

As an efficient catalyst, the interwoven network and high specific surface area of carbon aerogels could increase the number of active sites. Moreover, compared with other catalysts, carbon aerogels exhibit wonderful contaminant adsorption and enrichment capacity, which extremely enhances the catalytic performance. Therefore, carbon aerogels are frequently used as photocatalysts or electrocatalysts for pollutant removal in water or air.[34,129,167–172] For aquatic environments, carbon aerogels are mainly used for organic pollutant oxidation and ion reduction. For example, Tang et al. [169] fabricated a compressible g-C_3N_4/GO aerogel with recyclability and used it as photocatalyst for degradation of dyes (methyl orange (MO) and methylene blue (MB)) and reduction of bromates. The results indicated that g-C_3N_4 was evenly dispersed over the GO by π–π stacking, and the addition of GO significantly increased the photocatalytic activity. The degradation rate of MO and MB (20 mg L^{-1}) was over 90 % in 40 min, and the conversion rate of bromate (250 mg L^{-1}) was over 80 % in 60 min visible-light illumination. After 5 repeats, the mass loss of the hybrid aerogel was below 4 %, and the photocatalytic activity was not reduced. The mechanism study demonstrated that the organic pollutants were adsorbed by π–π interaction and predominantly degraded by peroxide radical anions during the photodegradation processes.

Zhang et al. [129] synthesized a polypyrrole/GO aerogel and used it as the electrode for simultaneous removal of hexavalent chromium and bisphenol A in aqueous solution. The influences of main parameters, including pH, concentration, voltage and the ratio of PGAs mass, were systemically investigated. They found that the max removal rates for hexavalent chromium and bisphenol A were 98.52 ± 1.48 % and 98.00 ± 1.17 % within 30 min. The electro-catalytic kinetics results show that the fitted value of K in mixed solution were significantly higher than that in one-component solution, demonstrating the synergy of both contaminants. The study provided a remarkable example of a "waste control by waste" strategy for the synergetic removal. Similarly, Jiang et al. [167] prepared a porous carbon aerogel with D-glucose, ammonium persulfate, and aniline and used it as a catalyst for activating persulfate (PS) to degrade rhodamine B (RhB). The PS concentration and CA dosage were more important than solution pH in the RhB degradation. The results confirmed that PS was activated by carbon aerogel in a non-radical way. RhB could be degraded by the PS bonded with C=C, and the removal efficiency was also facilitated by the defective edges at the boundary of carbon aerogels.

For air purification, carbon aerogels are mainly considered as photocatalysts. Hu et al. [34] used a graphitic-carbon nitride (g-C_3N_4) aerogel, which was modified with a perylene imide (PI) and graphene oxide, as a visible-light photocatalyst for nitric oxide (NO) removal. PI-g-C_3N_4 aerogel exhibited a strong light absorption, and graphene oxide was beneficial for the charge transportation and the diminution of electron–hole pairs recombination. Therefore, the functionalized aerogel exhibited an excellent photocatalytic activity in NO purification with a removal ratio up to 66 %. The possible mechanism for NO removal was proposed, as shown in Figure 8. The absorption of visible-light by the aerogel resulted in the formation of electron–hole pairs, and the electrons were excited to the conduction band (CB). Then, the electrons in the CB of PI transferred to the valence band (VB) of g-C_3N_4 and recombined with holes. Finally, NO was oxidized to NO^{3-} by the holes remaining in the VB of PI. Simultaneously, oxygen was quickly reduced to $\cdot O^{2-}$ and $\cdot OH$ by the excited electrons in the CB of g-C_3N_4, and NO was transformed to NO^{3-} with the intermediate product of NO_2.

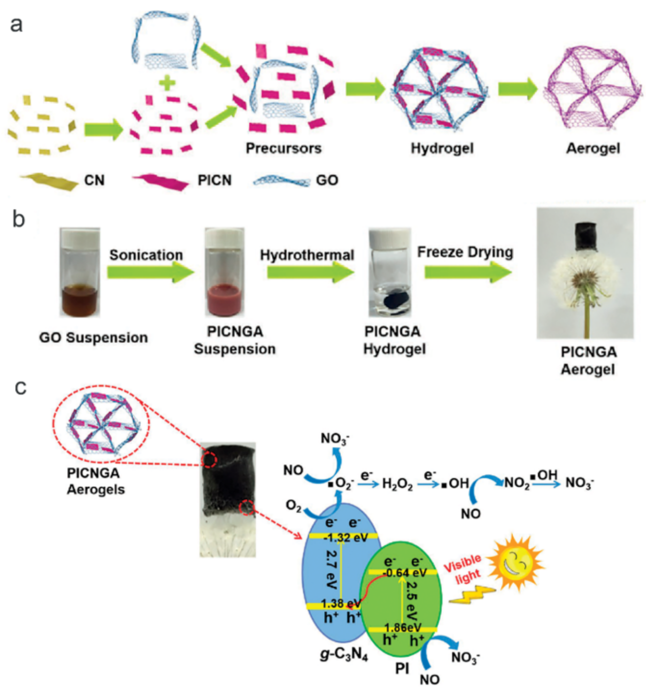

Figure 8. (a) Schematic illustration of the synthesis of the PICNGA composite aerogel. (b) Photographs of the aerogel fabrication process. (c) Photocatalytic mechanism of NO removal under visible-light irradiation by PICNGA aerogels. Reprinted from ref.[34]

4.2 Use as Catalyst Support

It is highly desirable for catalysts to use a support material to increase the activity. Coincidentally, carbon aerogels exhibit fascinating textural and structural properties, and the texture, composition, homogeneity as well as structural features of carbon aerogels can be flexibly controlled on a molecular level by the sol–gel technique. Various oxides or metals can be dispersed in the carbon aerogel matrix to design and tailor the chemistry toward the catalytic reactions. Therefore, carbon aerogels are frequently studied as an outstanding support material for catalyst .[28,29,32,90,112,139,173–190] For example, Liu et al. [174] fabricated $CeVO_4$/3D reduced graphene oxide aerogel/$BiVO_4$ ($CeVO_4$/RGO/$BiVO_4$) ternary composites and used them as photocatalysts for tetracycline (TC) degradation under visible light irradiation. The vanadates particles were surrounded by graphene aerogel networks, formed an organized heterostructure. And the photocatalytic activity was highly enhanced by the illuminated Z-scheme $CeVO_4$/RGO/$BiVO_4$ configuration. They found that RGO aerogel in the heterostructure acted as an effective electron bridge between two semiconductors and improved the charge transfer efficiency. Moreover, the research demonstrated that carbon aerogels were an ideal electron mediator to fulfil efficient electron-hole separation and charge transportation in Z-scheme semiconductor photocatalytic systems.

Electrochemical advanced oxidation processes (EAOP), as a new promising technique for removal of toxic and persistent organic pollutants, has received tremendous attention in recent years.[191] To improve the electrocatalytic activity, various carbonaceous materials are used to fabricate electrode, such as activated carbon, graphite felt and carbon aerogel. Among them, carbon aerogel, with a 3D monolithic network structure, high surface area, and good electrical conductivity, is an excellent choice for electrode. For example, Zhao et al. [28] designed a MOF(2Fe/Co)/carbon aerogel cathode for the solar photo–electro–Fenton (SPEF) process. The cathode was proved to have a good electrocatalytic and photocatalytic activity for oxygen reduction reaction (ORR), and the H_2O_2 was generated continuously over the composite cathode. The removal rate of Rhodamine B and dimethyl phthalate were 100 % and 85 % in the reaction time, respectively. In SPEF process, the application of carbon aerogel increased the active sites and hence enhanced the yield of the H_2O_2. At the same time, the H_2O_2 was efficiently decomposed by the photoinduced electron to form hydroxyl radicals. And it retained a high electrocatalytic and photocatalytic activity in the wide pH range of 3–9. Moreover, the iron and cobalt leaching were low due to the wonderful adsorption capacity of carbon aerogel.

Catalytic reduction to useful carbon-based fuels is an attractive approach to eliminate CO_2. Thus, it is necessary to develop a catalyst with high catalytic activity. In the recent years, metals or metal oxides supported on carbon aerogels have been recognized as fascinating electrocatalysts for the electrochemical reduction of CO_2. For example, Chen et al. [32] used carbon nanotube aerogel supported Sn spheroidal particles on carbon cloth (Sn/CNT-Agls/CC) electrode to selectively reduce CO_2 to formate. The electrocatalytic activity of Sn/CNT-Agls/CC was much higher than that of carbon nanotube-supported Sn spheroidal particles on carbon cloth (Sn/CNT/CC) electrodes, indicating that the physicochemical properties, including the 3D hierarchical structure, the high specific surface area, and the superior conductivity, were beneficial to the electrocatalytic activity. Observably, the selective formation of formate could be discovered at a low overpotential of 0.361 V, and the current density of Sn/CNT-Agls/CC electrode was 32.9 mA cm^2. Furthermore, the faradaic efficiency of formate production was as high as 82.7 % at 0.96 V.

As with CO_2 reduction, carbon aerogel supported catalyst was frequently used for CO oxidation. Qu et al. [188] loaded noble metal Ru into graphene aerogel to form a macroscopic 3D Ru/graphene aerogel/ HKUST-1 (Ru/GA/HK) CO oxidation cata-

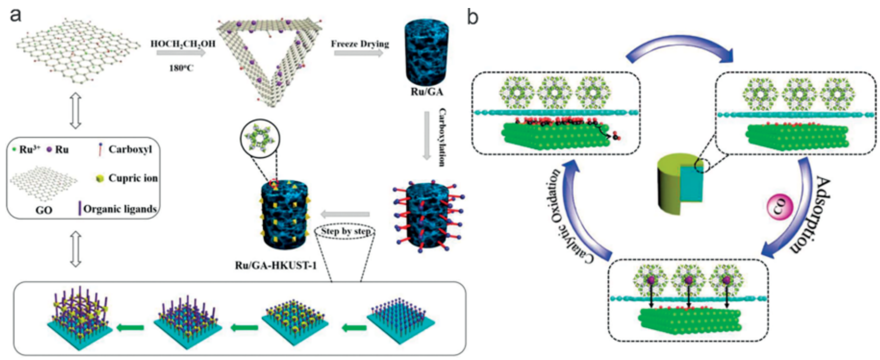

Figure 9. (a) Schematic illustration of the fabrication of Ru/GA-HK. (b) The reaction mechanism for CO removal. Reprinted from ref.[188]

lyst (Figure 9). Graphene aerogel brought open pathways for the access and diffusion of reactant and product molecules. Thus, CO could be simultaneously adsorbed and oxidized, resulting an enhanced reaction rate, and CO conversion rate of Ru/GA/HK catalyst was 100 % at room temperature. These findings further proved that the reaction rate could be increased by increasing the instantaneous concentration of reactants around the catalyst, and the strategy has potential application in air purification.

Hydrogen sulfide (H_2S), which is mainly eliminated from coal combustion, is another important pollutant in air environment. Hot coal gas desulfurization (HGD) process is one of the most efficient control techniques. carbon aerogels are considered as an attractive support for HGD catalysts due to their excellent physicochemical property as well as their stability under high temperature and no need for any other pretreatment. Tian et al. [189] fabricated a desulfurizer with Fe nanoparticles and carbon aerogels and investigated its H_2S removal efficiency in hot coal gas. The characterization results indicated that Fe NPs were highly dispersed on the surface of carbon aerogels, and the resultant Fe/CA-700 exhibited a high sulfur adsorption capacity of 12.54 g/100 g in hot coal gas under the 600 °C. Moreover, the Fe/CA-700 desulfurizer performed good antihydrogen ability, and the existence of reduction H_2 in hot coal gas had little effect on the desulfurization.

As can be seen above, carbon aerogels exhibit high adsorption capacity for VOCs, and we all know that the catalytic activity can be increased by increasing the instantaneous concentration of reactants. Therefore, carbon aerogels are suitable as a support for catalyst of VOCs catalytic combustion. Maldonado–Hódar [139] impregnated or doped Pt particles into carbon aerogels, the as prepared Pt-catalysts were applied to remove aromatic or oxygenated VOCs (toluene, xylenes, and acetone) by adsorption and catalytic combustion. Adsorption and degradation occurred simultaneously in the combustion process, and the lower temperature promoted the adsorption process while the catalytic reaction was favored at a higher temperature. Furthermore, they found that the combustion of aromatic compounds needed a lower temperature than that for acetone. The products of coke, CO or intermediate oxidized organic compounds were not detected except CO_2, indicating a high catalytic performance and CO_2 selectivity.

5. Conclusion and Outlook

Carbon aerogels are recognized as fascinating materials for environmental clean-up due to their unique and adjustable physical properties, including low apparent density, large specific surface area, abundant pore structure, high electrical conductivity, good chemical stability, environmental compatibility, as well as modifiable surface chemistry and controllable texture structural features. In this regard, the current advances in the research of these attractive materials applied for environmental protection are thoroughly reviewed in this paper. Up to now, carbon aerogels have mainly been used for pollutant removal based on their adsorption and catalytic performance.

As excellent sorbents, carbon aerogels are widely used both in water treatment and air purification, for applications like selective oil/water separation, adsorption of heavy metal ions as well as removal of volatile organic compounds (VOCs), CO_2 capture, and so on. In most studies, carbon aerogels have been found to have a higher sorption capacity towards the target components than other adsorbents, and they can be easily regenerated through squeezing, combusting, eluting, etc. These aspects make it possible to produce and apply carbon aerogels on a large scale.

In the catalytic field, carbon aerogels not only act as highly active catalysts, but are also used as supports for metal or metal oxide catalysts. For a catalyst, the 3D network structure increases the active sites, and the adsorption capacity increases the reaction concentration of contaminants, resulting in an enhanced catalytic activity. Moreover, the conductivity increases the mass transfer rate. Therefore, carbon aerogels are mainly used as photocatalysts and electrocatalysts in environmental protection. As catalyst supports, carbon aerogels fulfil all the necessary conditions. Their excellent compositional and structural properties can be easily regulated on the molecular level in the sol–gel process. Furthermore, the precursors of carbon aerogels exhibit high affinity to most catalysts, reducing the difficulty of catalyst loading.

Although carbon aerogels exhibit so many superiorities, there are lots of challenges for practical applications. Firstly, the synthetic processes of carbon aerogels are quite lengthy, and most of the precursors are expensive, resulting in high costs of production. Secondly, the high capillary tension causes the structural shrinkage and collapse under ambient pressure drying. To avoid that, supercritical drying and freeze-drying techniques are applied in many cases, increasing the operational risk and cost during preparation. Thirdly, many pristine carbon aerogels are fragile, weakening the regeneration of sorbent, especially for various water treatment purposes. Finally, the micromorphological features of a large amount of carbon aerogels are amorphous, which leads to the limitation of their selectivity in some applications.

All in all, much challenging work is still in front of us for the fabrication of carbon aerogels. Additional studies are still needed to: 1) decrease costs of carbon aerogel production by developing low-cost precursors and reducing the number of steps in the production processes; 2) overcome the capillary tension under ambient pressure drying and prevent the structural shrinkage and collapse; 3) increase the mechanical strength with new strategies; 4) further improve adsorption and catalytic capacity of carbon aerogels; 5) exploit new methodologies to regulate the microstructure and further enhance the selectivity; 6) expand the application fields, for example adsorption of particulate matter (PM2.5) and noise abatement.

Acknowledgments

This work was financially supported by the Key Project of the National Ministry of Science and Technology (No. 2016YFC0204200), the National Natural Science Foundation of China (No. 21577012), the Major Program of the National Natural Science Foundation of China (No. 21590813), the Program of Introducing Talents of Discipline to Universities (B13012), and the Key Laboratory of Industrial Ecology and Environmental Engineering, China Ministry of Education.

Keywords: Gels · Synthesis design · Environmental chemistry · Adsorption · Heterogeneous catalysis

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Received: December 14, 2018

Carbon Aerogels

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Carbon Aerogels for Environmental Clean-Up

Carbon aerogels have excellent properties including low apparent density, large specific surface area, abundant pore structure, high electrical conductivity, good chemical stability, environmental compatibility, as well as modifiable surface chemistry and controllable texture and structural features. Their applications as adsorbents and catalysts for environmental clean-up have been reviewed.

DOI: 10.1002/ejic.201801512

